Tax Evasion and Productivity

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Tax evasion through input overreporting

- VAT allows firms to deduct taxes paid on their purchases from the taxes collected on sales
- By overreporting their inputs, firms can reduce their tax bill
- Input overreporting is widespread: a common practice and a concern among OECD countries (OECD 2017)
  - Losses of 1.2–2.7% of total tax revenue in Chile, Colombia, Ecuador, Poland and Slovakia
  - Conditional on detected evasion: true losses are plausibly larger
- Governments have limited resources, and overreporting costs the government tax revenue

Question: Can we detect and measure tax evasion through input overreporting? How much does overreporting of inputs cost the government?

Why is it difficult to detect?

- Overreporting is unobserved: firms conceal their actions because it is illegal
- Tax-evading firms might *pass* as low-productivity firms
- For a given level of output, high input use by a firm might be because the firm is overreporting or because its productivity is low

What have people done?

- **Transaction-level data and administrative records** (Carrillo et al. 2022; Zumaya et al. 2021)
  - False invoicing, ghost-firm invoices, invoice mills
  - Access to this data is restricted; it is conditional on detected evasion
- **On the consumer side:** a benchmark group to measure misreporting from potentially evading individuals (Pissarides and Weber 1989; Gorodnichenko et al. 2009)
  - Self-employed vs. employees
  - The self-employed might underreport their income to pay less income tax; employees cannot misreport because their income is reported by their employers
  - Consumption is a function of true income: comparing their consumption backs out how much the self-employed underreport

The natural analogue for firms: output

- Output is a function of true inputs (assuming output is correctly reported)
- If we knew the production function (PF) and productivity, we could back out true inputs: the difference between reported and true inputs is the overreporting
- However, we rely on *reported inputs* to recover the PF and productivity
  - Proxy-variable methods rely on the *flexible* input (Levinsohn and Petrin 2003; Ackerberg et al. 2015; Gandhi et al. 2020)
    - If the flexible input is deductible, firms might overreport it: materials, in my application
- We have to solve the problem simultaneously, for overreporting and productivity

What do I do?

- Instead of looking into the PF equation, I take the FOC of the firm's profit-maximization problem
- The FOC links the materials share of revenue to the materials elasticity of output, up to measurement error
  - It does not depend on productivity (Hicks-neutral productivity)
- Just like in the individuals' case, I estimate this elasticity from a benchmark group of firms that do not overreport
  - The benchmark group can have different productivity than the potentially tax-evading firms
  - The critical assumption: both share a common technology
- With the elasticity at hand, the difference between reported and true materials reveals the overreporting, up to measurement error

What do I find?

Colombian manufacturing, 1981–1991:

1. **Detect:** overreporting in 9 of the 18 industries liable for the sales tax (about two fifths of manufacturing output); none in the 2 exempt food industries
2. **Measure:** potentially tax-evading firms overreport on average 4.5–23.1% of their true materials, depending on the industry
3. **Cost:** undetected overreporting costs 4.7% of gross sales-tax revenue; a 1% increase in the rate on purchases lowers net sales-tax revenue by about 1.7% of gross sales-tax revenue
4. **Productivity:** failing to correct for overreporting leads to biased estimates of the PF and productivity

Setting: Colombia, 1981–1991

- Annual Survey of Manufacturing (EAM): a panel of manufacturing plants with more than 10 employees
  - Output and inputs, juridical organization, and the sales tax charged on sales and paid on purchases
- The Colombian sales tax worked as a VAT: manufacturers credited the tax paid on their purchases against the tax charged on their sales
- The credit rewards overreporting materials: each fictitious peso lowers the tax liability by the rate on purchases
  - Capital goods could not be credited and labour was not a taxed purchase
- Weak enforcement: tax evasion flourished, and the tax authority audited inefficiently (Sánchez Torres and Gutiérrez Sourdis 1994; Perry and Orozco de Triana 1990)
- Food products (311, 312) were exempt from the sales tax: almost no reward to overreporting

The benchmark group: corporations

- Two kinds of scrutiny restrained corporations from overreporting: government and market
  - Government: the Superintendent of Corporations closely supervised corporations and required each to have an auditor (Joint Tax Program of the Organization of American States and the Inter-American Development Bank 1965)
  - Market: tradable shares subject corporations to market discipline; the market penalizes aggressive tax behaviour even when it is legal (Hanlon and Slemrod 2009)
- Limited liability companies (LLCs) and proprietorships faced only the tax authority, whose audits were weak
- Ecuador: tax evaded through fictitious purchases was 1.7% for corporations and 11.5% for sole proprietorships (Carrillo et al. 2022)
  - If corporations nonetheless overreport, my estimates are a lower bound
- Corporations have different productivities than the rest of the firms. But that's OK: the critical assumption is a common technology

Model: firms, taxes and overreporting

- Price-taking firms maximize expected after-tax profits; they share a common technology but differ in productivity

$$Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it} + \varepsilon_{it})$$

  - ω_{it}: productivity, known to the firm; ε_{it}: measurement error in output
- Materials are flexible and deductible; firms report $M_{it}^{*} = M_{it} + e_{it}$, $e_{it} \geq 0$
  - Overreporting is detected with a probability $q(e)$ and costs $\kappa(e, \omega)$; both are zero for a firm that does not overreport

Model: the firm's problem

$$\max_{M,e,K,L} \mathbb{E}[\pi_{it}] = (1-\tau_S)P_t E_\varepsilon[K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it}+\varepsilon_{it})] - (1-\tau_P)\rho_t M_{it} + \tau_P \rho_t e_{it}[1-(1+\phi)q(e_{it})] - \kappa_{it}(e_{it},\omega_{it})$$

- Revenue net of the tax on sales; the cost of true materials net of the credit on them; the rental cost of capital and the wage bill
- The credit on overreported materials net of the expected repayment and fine; the cost of evasion
- Underreporting would increase the tax liability: no firm has an incentive to do so, $e_{it} \geq 0$

Model: the materials FOC

- The FOC with respect to materials gives the log share of materials in revenue, net of sales taxes:

$$\ln\left(\frac{(1-\tau_P)\rho_t M_{it}}{(1-\tau_S)P_t Y_{it}}\right) = \ln\beta - \varepsilon_{it}$$

  - Productivity does not appear in the share (Hicks-neutral productivity)
- With reported materials M^*_{it}, the log share s_{it} satisfies

$$s_{it} = \ln\beta - \varepsilon_{it} + u_{it}, \qquad u_{it} = \ln(M^*_{it}/M_{it}) \geq 0$$

Testing for overreporting

- Corporations do not overreport: $s_{it} = \ln \beta - \varepsilon_{it}$ gives $\ln \beta = E[s_{it} \mid \text{corporations}]$
- Unincorporated firms: the observed residual $\mathcal{V}_{it} = s_{it} - \ln \beta = u_{it} - \varepsilon_{it}$

$$H_0 : \mu_{\mathcal{V}} = 0 \qquad \text{vs.} \qquad H_1 : \mu_{\mathcal{V}} > 0$$

- One-sided: no firm has an incentive to underreport deductible inputs. A rejection can only point to overreporting
- The test does not depend on the functional forms of the detection probability and the cost of evasion, nor on the Cobb-Douglas form of the technology
- Confidence set for $\mu_{\mathcal{V}}$ by test inversion, clustered by plant

Testing results

Industry	$\hat{\mu}$	95% region	Industry	$\hat{\mu}$	95% region
311 Food products	0.024	$(\cdot,\ 0.048]$	**342 Printing and publishing**	0.074	[0.036, 0.112]
312 Other food products	−0.018	$(\cdot,\ 0.076]$	**351 Industrial chemicals**	0.165	[0.001, 0.329]
313 Beverages	0.390	[0.310, 0.470]	**352 Other chemicals**	0.057	[0.002, 0.111]
321 Textiles	0.151	[0.115, 0.188]	356 Plastic products	−0.107	†
322 Wearing apparel	0.224	[0.195, 0.253]	**369 Non-metallic minerals**	0.193	[0.109, 0.277]
323 Leather products	−0.192	†	381 Metal products	−0.005	$(\cdot,\ 0.025]$
324 Footwear	0.095	[0.059, 0.131]	382 Non-electrical machinery	−0.014	$(\cdot,\ 0.026]$
331 Wood products	0.250	[0.175, 0.325]	383 Electrical machinery	0.034	$(\cdot,\ 0.089]$
332 Furniture	−0.007	$(\cdot,\ 0.059]$	384 Transport equipment	−0.129	†
341 Paper products	−0.066	$(\cdot,\ 0.004]$	390 Other manufacturing	0.057	$(\cdot,\ 0.115]$

$\hat{\mu}$: mean of $\mathcal{V}_{it}$ among unincorporated firms. 95% test-inversion region (sharp test); bold: the region excludes 0 (overreporting detected). $(\cdot,\ b]$: 0 not rejected. †: empty region.

- The test detects overreporting in 9 of the 18 industries liable for the sales tax: 4 of the 5 largest by output; 41% of manufacturing output

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- No overreporting in either exempt industry, Food products (311) and Other food products (312): this is expected

Counterfactual: the rate on purchases

- How does net sales-tax revenue change as the government changes the sales-tax rate on purchases?
- Expected net sales-tax revenue from firm i at the new rate $\tilde{\tau}_P = (1+\Delta)\tau_P$:

$$R_i(\Delta; M_i) = \underbrace{\tau_S P_t Y_{it}(\Delta)}_{\text{gross sales-tax revenue}} - \underbrace{\tilde{\tau}_P\left[M_{it}(\Delta) + (1 - q(\tilde{e}_i(\Delta)))\tilde{e}_i(\Delta)\right]}_{\text{claimed deductions}}$$

- A change in the rate moves net sales-tax revenue through three channels:
  1. Mechanical: every peso the firm claims is credited at the new rate
  2. Overreporting: a higher rate raises the marginal benefit of overreporting
  3. Materials: the rate changes the net-of-tax price of materials relative to output

The overreporting response: the evasion condition

- To compute the undetected overreporting at the new rate, $(1 - q(\tilde{e}_i))\tilde{e}_i$, we need the firm's FOC with respect to e:

$$\underbrace{\tau_P \rho_t \left[1 - (1+\phi)\left(q(e_{it}) + q'(e_{it})e_{it}\right)\right]}_{\text{marginal benefit}} = \underbrace{\frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}}}_{\text{marginal cost}}$$

- A higher tax rate on purchases raises overreporting
- The cost of evasion depends on productivity, which lets the model match the observed pattern in recent evidence (Carrillo et al. 2022)
- Estimating $q(e)$ and $\kappa(e, \omega)$ requires the latent true materials

Estimation: ELVIS

I estimate the model with ELVIS, Entropic Latent Variable Integration via Simulation (Schennach 2014)

What is ELVIS?

- A method of simulated moments where we use Markov chain Monte Carlo (MCMC) methods to sample from a non-parametric distribution of the unobservables conditional on the observables, such that the moment conditions hold
- It turns an existence question into an optimization problem: instead of asking “is there a distribution of the unobservables conditional on the observables such that the moments hold?”, it asks “what is the distribution of the unobservables conditional on the observables that maximizes entropy?”
- Why do we care about entropy? We don’t. We just need a distribution that satisfies the moment conditions
- ELVIS is always going to find a distribution: we test

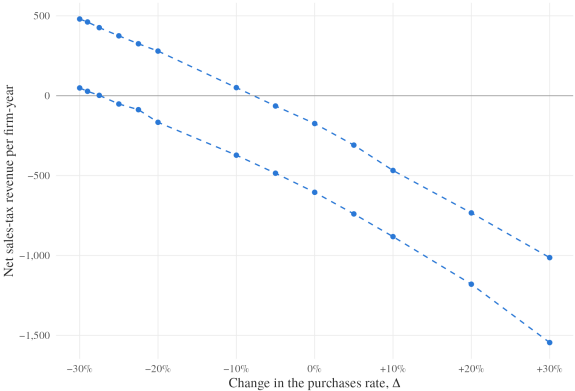
Why ELVIS?

- Moments are nonlinear in the unobservable, true materials. So a standard GMM or instruments/2SLS would not work
- We could assume a parametric distribution (indirect inference, maximum simulated likelihood). ELVIS does not impose parametric assumptions on the distribution of the unobservables

The counterfactual as a moment

- If we can write our counterfactual as a moment condition, then we can use ELVIS to answer the counterfactual
- Add the counterfactual moment, with an auxiliary parameter T, to the set of moments: $E\left[R_i(\Delta; M_i) - T\right] = 0$
- For each T: is there a distribution of the unobservables conditional on the observables that satisfies the model's moments AND the counterfactual moment? Yes or no
  - The confidence set collects the values of T that the test does not reject
- Why a set: the entropy-maximizing distribution should not be treated as the actual distribution of M, since many conditional distributions of M given Z may satisfy the moment condition (Schennach 2022)
  - "Latent variable models are often set-identified" (Schennach 2014, 346)

Results: net sales-tax revenue



Real pesos per firm-year, unincorporated firms in the 9 industries where the test rejects

- Current rate: -391 $[-605, -174]$: refunds on claimed deductions exceed gross sales-tax revenue
- Breaks even at a cut of about 17%; significantly positive from a cut of 27.5% $[2, 426]$

Results: the cost of overreporting

- Elasticity: A 1% increase lowers net sales-tax revenue by about 1.7% of gross sales-tax revenue $[1.47, 1.91]$
- Cost:
  - Undetected overreporting costs 4.7% $[2.8, 7.1]$ of gross sales-tax revenue from the unincorporated firms in the 9 industries
  - At least 1.5% $[1.1, 3.1]$ across all firms in the sample
  - Same order of magnitude as what tax authorities have detected from input overreporting: 1.2–2.7% of total tax revenue in five countries (OECD 2017; Carrillo et al. 2022; CASE and IHS 2018; Longinotti 2022)

Conclusion

- Can we detect and measure tax evasion through input overreporting? How much does it cost the government?
- Overreporting can be detected and measured from production data and a benchmark group alone
- It is common and large
- It is costly in two ways: for the government, which credits fictitious purchases, and for productivity measurement
- A higher rate on purchases raises claimed deductions by more than its mechanical effect:
  - net sales-tax revenue falls by more than the higher credit alone implies

Appendix

One unobservable: true materials

- I cast the estimation problem in terms of a single unobservable, true materials M:

$$e(M) = M^*_{it} - M, \qquad \varepsilon(M) = \ln(M^*_{it}/M) - \mathcal{V}_{it},$$

$$\omega(M) = \tilde{\mathcal{W}}_{it} - (1-\beta)\varepsilon(M)$$

  - $\mathcal{V}_{it}$: the observed residual of the test; $\tilde{\mathcal{W}}_{it}$: from the production-function estimates
  - The evasion condition then gives the cost shock $\psi(M)$
- Moments from the model's independence assumptions, e.g. $E[\psi] = 0$, $E[\psi \ln \tau_P] = 0$, $E[\varepsilon] = 0$

ELVIS: existence and optimization

- Theorem 2.1 of Schennach (2014): the existence problem is equivalent to an optimization problem

$$\inf_{f_{M|Z}} \left\| E[g(M,Z,\theta)] \right\| = 0 \iff \inf_{\gamma} \left\| E[\tilde{g}(Z,\theta,\gamma)] \right\| = 0,$$

$$\tilde{g}(Z,\theta,\gamma) = \frac{\int g(M,Z,\theta)\, e^{\gamma' g(M,Z,\theta)} f^0_{M|Z}(M|Z)\, dM}{\int e^{\gamma' g(M,Z,\theta)} f^0_{M|Z}(M|Z)\, dM}$$

- Among the conditional distributions of M given Z that satisfy the moment conditions, ELVIS selects the one with maximum entropy relative to a reference density $f^0_{M|Z}$; γ is the Lagrange multiplier on the moment constraints
- Test: $TS_n = 2n\hat{L}_n(\theta)$ against $\chi^2_{d_g}$ (conservative, Theorem F.1)

Model estimates

	Estimate
λ_1	0.75
$\bar{\lambda}$	0.50
$\hat{\delta}_0$	43.132
$\hat{\delta}_1$	27.149
$\hat{\delta}_2$	3.804
$\hat{\omega}^* = \hat{\delta}_1/2\hat{\delta}_2$	3.568
Firm-years (n)	12,050
Moment rows (d_g)	17
Directions kept	17 of 17
max $\lvert\hat{\gamma}\rvert$	195.4
TS_n ($R = 1000$)	23.69
TS_n ($R = 4000$)	24.74
$\chi^2_{17,.95}$	27.59
Result	**Passes**

Interior firms in the nine industries where the test rejects. Detection $q = (\lambda_{0,jt}e)^{-\lambda_1}$, $\lambda_{0,jt} = 1/(\bar{\lambda}\bar{M}_{j,t-1})$, with $(\lambda_1, \bar{\lambda})$ fixed at the operating point; R = draws per firm.

- Interior firms in the nine industries where the test rejects; the test statistic TS_n fails to reject at the 5% level
- Cost of evasion convex in productivity: $\hat{\delta}_1 = 27.15 > 0$, $\hat{\delta}_2 = 3.80 > 0$; it falls with productivity up to $\hat{\omega}^* = 3.57$ (about the 57th percentile) and rises after it
- Mean detection probability 6.8% $[5.7, 11.0]$; median 2.0%, 80th percentile 18.5%, 90th percentile 27%, maximum 57%
- The model's mean overreporting $E[u]$ is within 0.02 of the data's $E[\mathcal{V}]$ in seven of the nine industries

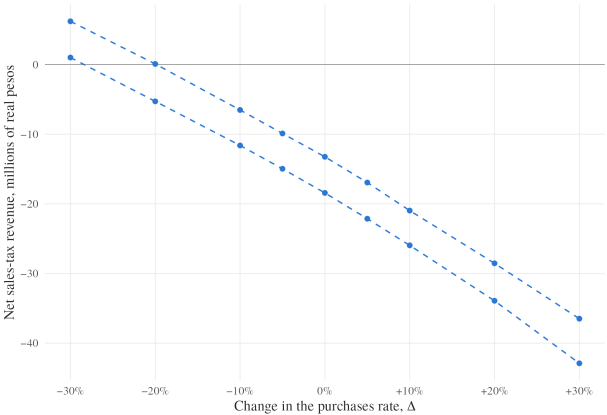
Claimed deductions: elasticities

	Claimed deductions			Net sales-tax revenue	
Δ	Total	Materials	Overreporting	Share of gross revenue	Relative percentage magnitude
-2%	1.29	0.17 [0.15, 0.20]	0.12 [0.10, 0.14]	-1.56 [-1.77, -1.38]	-6.19 [-7.02, -5.45]
-1.5%	1.31	0.18 [0.15, 0.20]	0.13 [0.10, 0.16]	-1.58 [-1.77, -1.39]	-6.23 [-6.98, -5.49]
-1%	1.32	0.18 [0.15, 0.21]	0.14 [0.12, 0.16]	-1.60 [-1.81, -1.38]	-6.32 [-7.16, -5.48]
$+1\%$	1.39	0.18 [0.15, 0.21]	0.21 [0.19, 0.23]	-1.68 [-1.91, -1.47]	-6.63 [-7.53, -5.81]
$+1.5\%$	1.40	0.18 [0.16, 0.21]	0.22 [0.20, 0.24]	-1.70 [-1.90, -1.50]	-6.71 [-7.51, -5.93]
$+2\%$	1.41	0.18 [0.15, 0.22]	0.23 [0.21, 0.24]	-1.71 [-1.93, -1.50]	-6.75 [-7.61, -5.94]

Arc elasticities with respect to the purchases rate, true materials responding, ELVIS interior firms; conservative 95% sets in brackets ($\chi^2_{18,.95} = 28.87$). Claimed deductions: $[C(\Delta) - C(0)]/(\Delta C(0)) = 1$ (mechanical) + materials response + overreporting response; the total is the sum of the point estimates. Net sales-tax revenue: share of gross revenue, $[R(\Delta) - R(0)]/(\Delta G(0))$, with mean gross sales-tax revenue $G(0) = 1{,}547$ real pesos per firm-year; relative percentage magnitude, $[R(\Delta) - R(0)]/(\Delta |R(0)|)$, with $|R(0)| = 391$.

- Claimed purchase deductions respond more than one for one: $+1\% \to 1.39 = 1$ (mechanical) $+ 0.18$ (materials) $+ 0.21$ (overreporting)
- At -1%: 1.32; the materials response is again 0.18, but the overreporting response falls to 0.14

Net sales-tax revenue: whole economy



Total over every firm-year in the sample, millions of real pesos

- Current rate: -15.9 $[-18.4, -13.2]$
- Breaks even at a cut of about 24%; a 30% cut raises it to 3.6 $[1.0, 6.2]$
- The whole economy needs a larger cut because of the corner firms in the other 19 industries

Robustness: true materials fixed

	True materials respond		True materials fixed	
Δ	Estimate	95% set	Estimate	95% set
−30%	264	[49, 480]	223	[0, 439]
−20%	59	[−166, 279]	23	[−180, 240]
−10%	−159	[−372, 51]	−183	[−395, 30]
−5%	−272	[−485, −64]	−285	[−486, −77]
0	−391	[−605, −174]	−391	[−605, −174]
+5%	−528	[−740, −309]	−512	[−709, −310]
+10%	−668	[−882, −468]	−640	[−840, −435]
+20%	−965	[−1,180, −733]	−899	[−1,126, −668]
+30%	−1,270	[−1,545, −1,013]	−1,163	[−1,386, −867]

Real pesos per firm-year, ELVIS interior firms (12,050 firm-years). Net sales-tax revenue = gross sales-tax revenue minus expected sales-tax refunds. 95% set: conservative test, $TS \leq \chi^2_{18,.95} = 28.87$. True materials respond: the materials first-order condition with the two tax rates, capital, labour and productivity fixed. True materials fixed: true materials stay at their current level. Break-even of the estimate, interpolating linearly: 17% cut with true materials responding, 19% cut with true materials fixed.

Test for overreporting: conservative test

Industry	$\hat{\mu}$	95% region	Industry	$\hat{\mu}$	95% region
311 Food products	0.024	$(\cdot,\ 0.054]$	**342 Printing and publishing**	0.074	[0.026, 0.122]
312 Other food products	−0.018	$(\cdot,\ 0.100]$	351 Industrial chemicals	0.165	$(\cdot,\ 0.375]$
313 Beverages	0.390	[0.287, 0.492]	352 Other chemicals	0.057	$(\cdot,\ 0.125]$
321 Textiles	0.151	[0.105, 0.197]	356 Plastic products	−0.107	†
322 Wearing apparel	0.224	[0.187, 0.261]	**369 Non-metallic minerals**	0.193	[0.087, 0.298]
323 Leather products	−0.192	†	381 Metal products	−0.005	$(\cdot,\ 0.033]$
324 Footwear	0.095	[0.050, 0.140]	382 Non-electrical machinery	−0.014	$(\cdot,\ 0.036]$
331 Wood products	0.250	[0.156, 0.344]	383 Electrical machinery	0.034	$(\cdot,\ 0.103]$
332 Furniture	−0.007	$(\cdot,\ 0.075]$	384 Transport equipment	−0.129	†
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$\hat{\mu}$: mean of $\mathcal{V}_{it}$ among unincorporated firms. 95% test-inversion region, conservative test ($\chi^2_{2,0.95} = 5.99$); bold: the region excludes 0. $(\cdot,\ b]$: 0 not rejected. †: empty region.

How much: deconvolution

Industry	Mean	SD	P10	Median	P90
321 Textiles	16.2%	5.3%	9.5%	15.8%	23.4%
322 Wearing apparel	23.1%	3.9%	18.0%	23.1%	28.2%
324 Footwear	10.5%	5.8%	4.4%	9.3%	18.2%
342 Printing and publishing	4.5%	7.2%	3.2%	4.0%	5.1%
369 Non-metallic minerals	12.4%	2.3%	9.5%	12.2%	15.4%

Overreporting ratio $x = e/M$: overreported materials as a share of true materials. Moments and quantiles of the density $f_x(y) = f_u(\ln(1+y))/(1+y)$, obtained by transforming the deconvolved density of $u = \ln(1+e/M)$ (penalized B-spline deconvolution, log materials share net of sales taxes). Sample: unincorporated firms that pay sales tax on purchases, the sample of the structural estimation in Chapter 8. Industries: the five largest, by share of output, where overreporting is detected at the 1% level and the variance of the unincorporated firms' residual is at least the corporations' (the condition the deconvolution requires; Beverages, 313, fails it) (all nine in Appendix E).

- In the five largest industries where the test rejects at the 1% level and the deconvolution applies, unincorporated firms overreport on average between 4.5 and 23.1% of their true materials

Production function estimates

		Corrected, $\hat{W}_{it-2}$		GNR			OLS		
Industry	$\hat{\beta}$	$\hat{\alpha}_K$	$\hat{\alpha}_L$	$\hat{\beta}$	$\hat{\alpha}_K$	$\hat{\alpha}_L$	$\hat{\beta}$	$\hat{\alpha}_K$	$\hat{\alpha}_L$
321	0.385	0.300	0.349	0.435	0.225	0.340	0.681	0.093	0.274
	[0.357, 0.416]	[0.210, 0.375]	[0.230, 0.485]						
322	0.335	0.182	0.392	0.417	0.175	0.343	0.578	0.047	0.397
	[0.288, 0.391]	[0.085, 0.265]	[0.320, 0.470]						
324	0.412	0.119	0.427	0.451	0.131	0.406	0.705	0.013	0.300
	[0.365, 0.465]	[−0.215, 0.270]	[0.190, 0.570]						
342	0.333	0.292	0.408	0.355	0.221	0.476	0.578	0.066	0.430
	[0.294, 0.377]	[0.215, 0.360]	[0.305, 0.530]						
369	0.244	0.333	0.549	0.279	0.254	0.557	0.495	0.128	0.519
	[0.207, 0.288]	[0.210, 0.485]	[0.210, 0.740]						

Evasion-corrected: $\hat{\beta}$ is the corporations' first-stage estimate (log materials share net of sales taxes); below it, its 95% test-inversion region from the corporations' share moment (χ^2_1). Point estimates of (α_K, α_L): the minimum of the test-inversion statistic, with the covariance of the moments re-estimated at each candidate, the statistic whose regions are shown. Instrument $\hat{W}_{it-2}$ (exactly identified); below the estimates, the projections of the 95% test-inversion region for (α_K, α_L) (χ^2_2, 0.005 grid over $[0, 1]^2$, extended beyond it where the region reaches its edge). Conservative regions: Appendix E. GNR and OLS: uncorrected estimates on the same sample (GNR's share regression on the log materials share net of sales taxes). Other instruments: Appendix E. Industries: the five largest, by share of output, where overreporting is detected at the 1% level and the variance of the unincorporated firms' residual is at least the corporations' (the condition the deconvolution requires; Beverages, 313, fails it) (all nine in Appendix E).

- The uncorrected estimates show output elasticities of materials that are consistently larger than the corrected ones

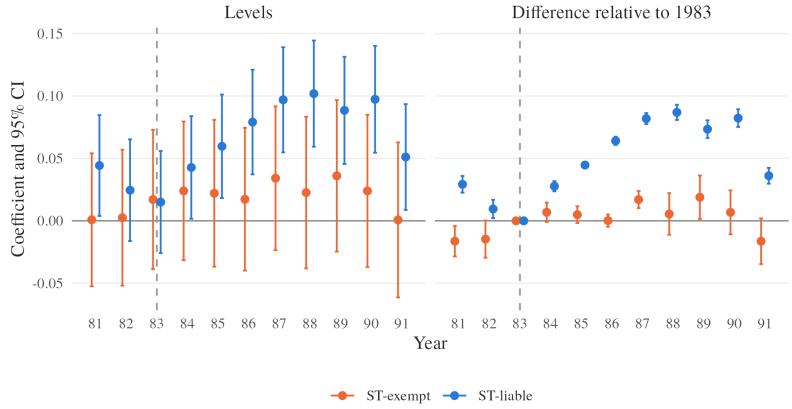
Productivity

Industry	Method	Mean	SD	Median	75/25	80/20	90/10	95/5	Skewness	$\hat{\gamma}_1$
321	Corrected	16.5	6.7	15.2	1.41	1.56	2.30	3.39	-0.04	0.95
	GNR	21.2	11.0	19.0	1.82	2.15	3.38	5.31	-0.34	0.87
322	Corrected	57.8	15.9	57.6	1.42	1.58	2.16	2.73	-0.48	0.94
	GNR	34.9	16.5	33.3	1.80	2.13	3.59	5.65	-0.70	0.88
324	Corrected	42.4	11.9	40.8	1.32	1.44	1.94	2.49	0.28	0.96
	GNR	28.8	10.4	27.5	1.53	1.74	2.38	3.08	-0.17	0.87
342	Corrected	19.1	4.0	19.0	1.25	1.33	1.70	2.13	-0.30	0.93
	GNR	24.2	11.5	22.2	1.69	1.92	2.88	4.14	-0.20	0.81
369	Corrected	19.0	5.6	18.6	1.30	1.39	1.75	2.41	-0.65	0.93
	GNR	32.8	22.9	28.3	2.69	3.33	6.86	10.95	-0.49	0.89

Productivity in levels, $\exp(\omega_{it})$; percentile ratios as in Gandhi, Navarro and Rivers (2020, Table 3); skewness of ω_{it}; $\hat{\gamma}_1$: persistence, the AR(1) coefficient of ω_{it} (corrected: IV estimate within the production function step; GNR: OLS on firm-level ω_{it}). Corrected: moments of the deconvolved density of ω (penalized B-spline deconvolution of $\widetilde{W}_{it} = \omega_{it} + (1-\beta)\,\varepsilon_{it}$), with the production function estimates of instrument $\widetilde{W}_{it-2}$. GNR: firm-level ω_{it} from the uncorrected GNR (2020) Cobb-Douglas estimation, with the measurement error removed. Both exclude ε. Log materials share net of sales taxes, juridical organization codes 6–9 excluded. Industries: the five largest industries, by share of output, where overreporting is detected at the 1 percent level and the variance of the unincorporated firms' residual is at least the corporations' (the condition the deconvolution requires; Beverages, 313, fails it).

- Uncorrected productivity distributions are more dispersed and less persistent: in the median of the five industries, the 90/10 ratio is 66% larger and the half-life of productivity shocks is 63% shorter

The 1983 reform



- Between 1983 and 1987, after Colombia raised the sales-tax rate for most manufacturing industries, overreporting by unincorporated firms in the affected industries rose by about 8.5 percentage points of true materials

The 1983 reform: elasticity

Year	Overreporting, 1983	Overreporting, year	τ_P, 1983	τ_P, year	Elasticity	95% set
1984	2.7%	5.6%	8.2%	9.3%	5.1	[2.2, 10.4]
1985	2.7%	7.4%	8.2%	10.2%	4.1	[2.4, 6.9]
1986	2.7%	9.4%	8.2%	10.2%	5.0	[3.3, 7.5]
1987	2.7%	11.3%	8.2%	10.0%	5.9	[4.2, 8.5]
1988	2.7%	11.7%	8.2%	9.8%	6.6	[4.7, 9.5]

Midpoint elasticity of overreporting, as a share of true materials, with respect to the sales tax rate on purchases τ_P, 1983 to each year: the change in overreporting over its midpoint, divided by the change in τ_P over its midpoint. Unincorporated firms in sales-tax-liable industries; τ_P columns: mean rate among these firms. The 95% set collects the elasticities that a χ^2_1 test does not reject at 5%, taking the corporations' mean log share by industry, the overreporting levels and the mean rates as the truth; variance clustered by plant. Overreporting cannot be negative, which caps the elasticity at 9.8 in 1987.

- The response implies a midpoint elasticity of overreporting with respect to the tax rate of 5.9

Overreporting elsewhere: Ecuador and Mexico

- **Share of true purchases.** Ecuador: among firms that use ghost receipts, ghost purchases are 10.4% of the value of their purchase deductions (3.6% of their purchases)
  - Mine: unincorporated firms overreport 4.5–23.1% of their true materials on average
- **Losses.** Detected losses 1.2–2.7% of total tax revenue in Chile, Colombia, Ecuador, Poland and Slovakia (OECD 2017; Carrillo et al. 2022; CASE and IHS 2018; Longinotti 2022)
  - Mine: 4.7% of gross sales-tax revenue (interior firms), 1.5% (all firms)
  - Theirs count only what enforcement detects: true losses plausibly larger
- **Corporations as benchmark.** Ecuador: tax evaded through fake receipts 1.7% for corporations vs 11.5% for sole proprietorships (Carrillo et al. 2022, 6)
- **Pattern by size.** Ecuador: share of deductions backed by fake receipts rises with revenue, drops sharply at the very top (Carrillo et al. 2022, 6)
  - Mine: cost of evasion convex in productivity, minimum at $\widehat{\omega}^* = 3.57$ ($\approx$ 57th percentile)

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